Universal File Extractor - Comprehensive Test Document

Document Metadata

This document contains extensive testing material for the Universal File Metadata & Version History Extractor. It includes multiple sections, paragraphs, and formatting to test extraction capabilities.

1. Project Overview

This comprehensive test document is designed to validate the Universal File Metadata & Version History Extractor's ability to process complex PDF files with substantial content. The document contains multiple paragraphs, sections, and formatting elements that will test the extraction system's robustness and accuracy. The primary objectives of this test are to verify metadata extraction, content parsing, and knowledge graph generation for PDF documents with extensive content. This ensures the system can handle real-world documents that contain detailed information across multiple pages.

2. Technical Specifications

The Universal File Metadata & Version History Extractor supports multiple file formats including PDF, Excel, Word, and plain text files. For PDF files, the extractor utilizes PyPDF2 and pdfplumber libraries to extract document metadata, text content, and embedded information. Key technical features include: • Multi-model extraction using multiple libraries per file type • Knowledge graph construction with nodes for each edit/change • Evaluation bands for confidence scoring (High/Medium/Low) • Version history tracking from multiple sources • Support for embedded markers, filesystem metadata, and native format history

3. Implementation Details

The extraction system follows a structured six-stage flow pipeline: 1. Ingestion - File detection and validation 2. Multi-Model Extraction - Apply all applicable extraction methods 3. Version Analysis - Extract edit history from available sources 4. Knowledge Graph Construction - Build graph of file evolution 5. Evaluation Band Assignment - Assign confidence scores 6. Output Generation - Produce JSON, GraphML, or GEXF format Each stage processes files consistently regardless of type, ensuring reliable results across different formats and content structures.

4. Testing Methodology

Comprehensive testing involves validating extraction accuracy across multiple file types and content complexities. Test files should contain known metadata, edit histories, and content patterns to verify correct extraction and knowledge graph generation. For PDF files specifically, tests validate: • Document metadata extraction (Title, Author, Subject, Creation Date) • Page count and structure analysis • Text content extraction accuracy • Knowledge graph node and edge creation • Confidence

scoring based on extraction source quality

5. Expected Results

When processing this comprehensive test document, the Universal File Extractor should: 1. Detect the PDF file type automatically 2. Extract all document metadata including title and author 3. Count pages accurately (this document has 2-3 pages) 4. Create a knowledge graph with file node and edit nodes 5. Assign high confidence scores based on metadata availability 6. Generate structured JSON output with all extracted information The extraction results should be consistent and complete, providing a full view of the document's metadata and structure.